

BOOK-STORE - Project Report

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Technology: Full Stack Development with MERN

Organization: APSCHE – Skill Wallet

GitHub Repository: <https://github.com/niharika-mente/BOOK-STORE>

Live Deployment URL: <https://bookverse-web.netlify.app/>

Project Video Demo:

https://drive.google.com/file/d/1MiSypwopk_WhKk1OztqkeSgXlkejzQOG/view?usp=sharing

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1. Abstract

BookVerse is a robust, full-stack bookstore management application developed leveraging the MERN stack ecosystem. The system provides seamless implementation for secure user authentication, interactive book discovery, customized user favourites tracking, shopping cart persistence, dynamic order management workflows, and an administrative control panel optimized for book and order fulfillment processing. The engineering focus is directed toward cross-device responsive design primitives, architectural scalability, and an intuitive user interface flow.

2. Introduction

BookVerse functions as an online bookstore marketplace designed to allow consumers to seamlessly discover, review, and acquire literary works through a streamlined user experience. By integrating MongoDB, Express.js, React, and Node.js components, this application serves as an end-to-end template for modern full-stack engineering. Key architectural pillars include secured identity validation layers, structured RESTful API layers, adaptive component sizing, and transactional database mapping

3. Project Overview & Features

User Features

- **Secured Identity Lifecycle:** Client sign-up and authentication procedures protected via JSON Web Tokens (JWT) and Bcrypt backend encryption layers.
- **Catalog Exploration:** Modular interfaces to search, filter, and inspect detailed book specifications.
- **Personalized Curation:** Direct entry pathways to manage user-specific favorite collections.
- **Transactional Cart Utility:** Real-time shopping cart calculation adjusting item amounts and price summations dynamically.
- **Order Tracking Pipeline:** Instant order creation paired with dedicated historic tracking views.
- **Profile Provisioning:** Dedicated panels allowing individual address validation and personal profile modification.

Admin Features

- **Inventory Control:** Complete administrative permissions to execute creation, modification, and deletion (CRUD) commands on books.
- **Fulfillment Management:** Complete logistical control over global customer order collections with state update capabilities.

4. System Architecture & Technology Stack

- **Frontend UI Framework:** React 19, Vite compilation engine, Redux Toolkit state storage, Tailwind CSS design utilities, React Router DOM, and Axios HTTP clients.
- **Backend Application Server:** Node.js runtime coupled with the Express.js routing engine.
- **Database Management Tier:** MongoDB database clusters orchestrated using Mongoose Object Data Modeling (ODM) layers.
- **Security & Cryptography Protocol:** JSON Web Token (JWT) stateless tokens paired with BcryptJS data salting.

- **Server Request Handling Middleware:** Cross-Origin Resource Sharing (CORS) configurations and built-in Express JSON parsers.

5. Project Structure

BOOK-STORE/

```
|— backend/
|   |— connections/  # Database configuration & Mongoose connection setup
|   |— models/       # Mongoose schemas (Book, User, Order)
|   |— routes/       # Express API endpoints grouped by controller
|   |— app.js        # Express server entry point
|   |— package.json  # Backend dependencies & scripts
|   |— .env          # Environment configurations
|
|— frontend/
|   |— src/
|   |   |— components/ # Reusable UI elements (Navbar, Footer, BookCard, Loaders)
|   |   |— pages/      # Views (Home, AllBooks, Profile, Cart, Signup, Login)
|   |   |— store/      # Redux store slices (Auth management)
|   |   |— App.jsx     # Main layout & router settings
|   |   |— main.jsx    # Vite entry point
|   |— package.json  # Frontend dependencies & configurations
|   |— vite.config.js # Vite configuration
|
|— README.md        # Project documentation
```

6.Setup & Installation Instructions

Follow these steps to set up and run the BookVerse application on your local development machine.

Prerequisites

Before starting, ensure you have the following installed on your system:

- Node.js (v16.x or higher recommended) & npm (Node Package Manager)
- Git (for cloning the repository)
- A MongoDB Atlas account (or a locally running MongoDB Community Server instance)

Step 1: Clone the Repository

Open your terminal (or Command Prompt) and run the following commands to clone the repository to your local machine and navigate into the root directory:

```
Bash

# Clone the repository
git clone https://github.com/niharika-mente/BOOK-STORE.git

# Navigate into the project root folder
cd BOOK-STORE
```

Step 2: Backend (Server) Configuration

1. Navigate to the server directory i.e backend using **cd backend**
2. Install all server-side dependencies using npm install: This reads the package.json file in your backend folder and installs Express, Mongoose, Bcrypt, JWT, and other required libraries.
3. Configure the Environment Variables (.env): Create a new file named .env in the root of your backend directory:

Open the .env file in your code editor (e.g., VS Code) and add the following configuration variables:

PORT=1000

URI=mongodb+srv://<username>:<password>@cluster0.mongodb.net/BookVerse?retryWrites=true&w=majority

JWT_SECRET=your_super_secret_jwt_key

Make sure to replace <username> and <password> with your actual MongoDB Atlas cluster credentials.

4. Start the Backend Server (Execute Server Daemon): Run the development script to boot up your Express API server: **npm run dev**

Keep this terminal window open. You should see a console message indicating that the server is running on your specified port and has successfully connected to MongoDB.

Step 3: Frontend (Client) Configuration

Crucial: Open a new terminal window or tab so your backend server can continue running undisturbed in the background.

1. Navigate to the frontend directory i.e frontend

```
Bash

# From the BOOK-STORE root directory:
cd frontend
```

2. Install all client-side dependencies using npm install: This installs React 19, Vite development tools, Redux Toolkit, Tailwind CSS, Axios, and React Router DOM.

Launch the Frontend Application: Start the Vite development engine locally using **npm run dev**

Accessing the Application

Once both servers are running successfully:

- Your Backend API will be accessible at: **http://localhost:1000**
- Your Frontend Application (Vite default server) will be accessible at: **http://localhost:5173** (or the local URL printed in your second terminal window). Open this URL in your web browser to interact with BookVerse!

7. API Endpoints

- **Identity Management Layer:** Handles user registration (Sign Up), session creation (Sign In), profile recovery (Get User), and logistical tracking (Update Address).
- **Inventory Catalog Route:** Handles creation routing (Add Book), alteration routing (Update Book), wipe routing (Delete Book), along with selective query hooks (Get All, Get Recent, Get By ID).
- **Curation Hooks (Favorites):** Direct endpoints allowing users to add to favorites (Add), eliminate from favorites (Remove), and retrieve lists (Get Favorites).
- **Cart Transactions:** REST operations allowing full manipulation of items within the active session cart (Add, Remove, Get Cart).
- **Fulfillment Pipeline (Orders):** Transaction systems managing checkout actions (Place Order), consumer logs (Order History), master administrator tracking (All Orders), and status flags (Update Status).

8. Authentication & Authorization Handling

Data Verification: Identity confirmation paths rely entirely on JSON Web Tokens (JWT) distributed upon successful verification, alongside BcryptJS hashing mechanisms preventing raw text leak vulnerabilities.

Role-Based Access Guardrails: Routing endpoints are secured at the middleware layer to strictly segment consumer interaction routes from privileged administrative database commands.

9. User Interface Design & Visual Map

Landing Page Display: Built with a minimal theme featuring clear marketing headers ("A World of Stories, One Click Away") and exploration action buttons

Catalog Exploration Matrix: Formatted card grids highlighting complete item information blocks like cover pictures, creator data, and regional currencies (e.g., *Emma* listed at ₹399; *The 48 Laws of Power* listed at ₹549)

Access Validation Portals: Specialized account forms featuring clean input structures for creation and credential matching dashboards ("Welcome Back")

Consumer Management Board: Tabbed profiles that let users interact cleanly with separate favorite tracking tables, chronological order logs, and settings tabs

Checkout Modules: Dedicated selection screens highlighting product descriptions, editable amount controls, language flags, and execution buttons ("Add to Cart", "Add to Favourites")

Master Administrative Interface: Centralized logging consoles ("All Orders") containing dynamic control forms to introduce new items or update inventory records

10. Testing Metrics

- **Access Control Verification:** Identity cycles, token storage, and session validation checks were tested successfully
- **Database Interaction Checks:** Full execution loops for data modification (CRUD) on the catalog schema were verified
- **Transaction Testing:** Checkout workflows from cart storage modifications to successful backend order placement were validated
- **Adaptive Optimization:** The application interface was systematically verified across varying viewports and desktop layouts to ensure cross-device consistency
- **Integration Reliability:** API responses and schema tracking inside the cloud database instances were verified

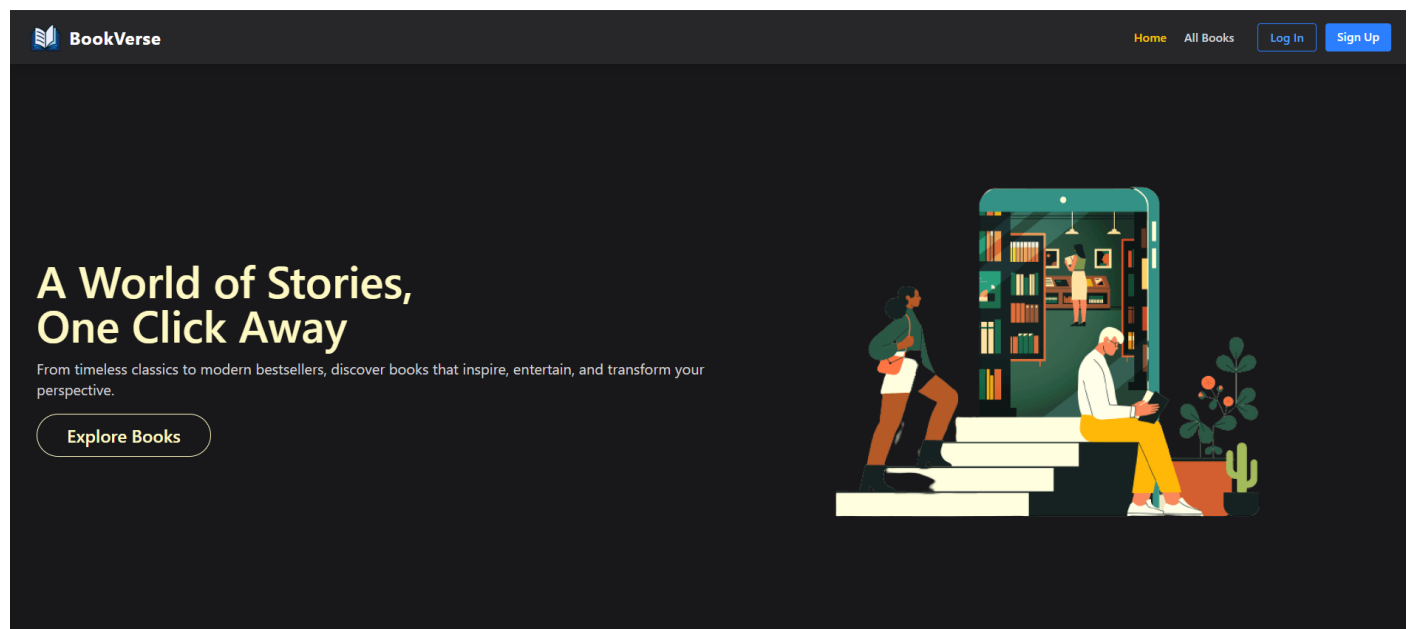
11. Known Issues

- *No critical bugs or blocking anomalies were recorded during system validation cycles.*

12. Results

The application was successfully developed and deployed with a responsive interface and secure backend. Users can browse books, manage favourites and carts, place orders, while administrators can efficiently manage books and customer orders.

HOME PAGE--



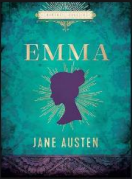
BOOKS PAGE--

BookVerse

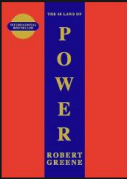
HomeAll BooksLog InSign Up

All Books

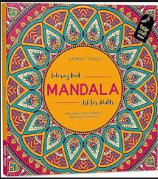
10 books available



Emma
by Jane Austen
₹399



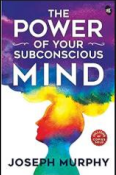
The 48 Laws of Power
by Robert Greene
₹549



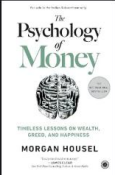
Mandala Art
by Various Authors
₹249



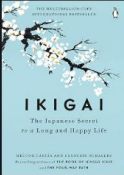
Twisted Love
by Ana Huang
₹399



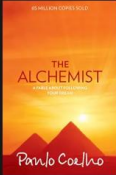
The Power of Your Subconscious Mind
by Joseph Murphy



The Psychology of Money
by Morgan Housel



IKIGAI
The Japanese Secret to a Long and Happy Life



THE ALCHEMIST
Paulo Coelho

SIGN-UP:

BookVerse

HomeAll BooksLog InSign Up

Create an Account

Join BookVerse and explore your next read

Username

Enter your username

Email

xyz@example.com

Password

Enter your password

Show

Must be more than 5 characters

Address

Enter your address

Sign Up

or

Already have an account? Log In

LOGIN PAGE:

BookVerse

HomeAll BooksLog InSign Up

Welcome Back

Log in to your BookVerse account

Username

user

Password

Show

Log In

or

Don't have an account? Sign Up

USER DASHBOARD:

BookVerse

HomeAll BooksCartProfile

user

abc1@gmail.com

Member

Favourites

Order History

Settings

Log Out

Order History

SR NO.	BOOK	DESCRIPTION	QTY	PRICE	STATUS
1	Ikigai	Ikigai explores the Japanese philosophy of living a lon...	2	₹798 ₹399 × 2	Out for Delivery
2	The Psychology of Money	The Psychology of Money explores how human emoti...	3	₹1260 ₹420 × 3	Delivered

ADD TO CART:

BookVerse

HomeAll BooksCartProfile

THE 48 LAWS OF

POWER

ROBERT GREENE

INTERNATIONAL BESTSELLER

♡

🛒

The 48 Laws of Power

by Robert Greene

🌐 English

₹549

About this book

The 48 Laws of Power is a bestselling self-help and strategy book that examines the timeless principles of influence, leadership, and human behavior. Drawing from historical events and the lives of influential figures, Robert Greene presents 48 practical laws that reveal how power is gained, maintained, and defended. Each law is supported by engaging stories, historical examples, and practical insights, helping readers understand social dynamics, decision-making, persuasion, and the complexities of personal and professional relationships.

Quantity

-

1

+

Add to Cart

Add to Favourites

✓ 1 × "The 48 Laws of Power" added to cart!

ADMIN DASHBOARD:

BookVerse

HomeAll BooksAdmin Profile

Admin

admin

admin123@gmail.com

Administrator

All Orders

Add Book

All Orders13

#	BOOK	USER	QTY	PRICE	STATUS	UPDATE
1	Ikigai	user	2	₹798 ₹399 × 2	Out for Delivery	Out for Delivery
2	The Psychology of Money	user	3	₹1260 ₹420 × 3	Delivered	Delivered
3	Mandala Art	test1	1	₹249	Cancelled	Cancelled
4	Twisted Love	test1	1	₹399	Delivered	Delivered
5	Ikigai	test1	2	₹798 ₹399 × 2	Delivered	Delivered
6	Twisted Love	test1	1	₹399	Order Placed	Order Placed
7	Book Deleted	ninu	3	₹0 ₹ × 3	Order Placed	Order Placed
8	The Power of Your Subconscious Mind	ninu	1	₹299	Order Placed	Order Placed

EDIT/DELETE BOOK BY ADMIN:



ADD NEW BOOK:

13. Future Scope & Enhancements

- Online payment integration.
- AI-powered book recommendations.
- Ratings and reviews.
- Email notifications.
- Admin analytics dashboard.
- Mobile application using React Native.

14. Conclusion

The development of **BookVerse** successfully demonstrates the practical application of modern, end-to-end full-stack web engineering principles. By leveraging the power and flexibility of the **MERN (MongoDB, Express.js, React, Node.js) stack**, the project achieves a seamless blend of high performance, security, and responsive design.

Key accomplishments realized through this development lifecycle include:

- **Robust Security Infrastructure:** Implementing state-of-the-art token-based authorization via **JSON Web Tokens (JWT)** and secure password salting with **BcryptJS** ensures that user profiles and transaction processes remain thoroughly protected.
- **Modular Frontend Architecture:** Utilizing **React 19**, **Vite**, and **Redux Toolkit** allows for highly scalable state management and near-instant page renders, elevating the user experience across both mobile and desktop viewports.
- **Efficient Database Mapping:** Designing clean schema architectures using **Mongoose ODM** optimizes database transaction handling for books, shopping carts, and dynamic customer order states within **MongoDB**.

As an initiative under the **APSCHE – Skill Wallet** program, this project bridges the gap between theoretical computer science concepts and real-world deployment practices, serving as a solid foundation for enterprise-level application development. Ultimately, **BookVerse** transitions from a concept into a fully functional, live-deployed e-commerce platform.